

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
H2 Biology Preliminary Examinations 2017

H2 BIOLOGY
 Practical

9744/04
02 Aug 2017
2 hours and 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, index number, class, shift and laboratory on this Question Paper.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	/21
2	/20
3	/14
Total	55

This question paper consists of 18 printed pages.

- 1** During anaerobic respiration, yeast cells use glucose and release carbon dioxide and ethanol. Ethanol is known to disrupt enzymes involved in the respiration pathway.

In this experiment, you will investigate the effect of different concentrations of ethanol on the rate of respiration in yeast cells.

- (a)** Sketch a fully-labelled graph to show the expected relationship between the rate of respiration and the concentration of ethanol, as ethanol concentration increases.

Explain the shape of your graph.

[4]

Methylene blue is a dye which, under certain conditions, is easily reduced to a colourless compound. In the presence of reduced NAD which is produced during glycolysis, methylene turns from blue to colourless.

In your investigation, you are to dilute the given ethanol solution to obtain different concentrations of ethanol solution.

You are provided with:

- 30 cm³ of 5.0% yeast solution, **Y**
- 20 cm³ of 100% ethanol, **E**
- 25 cm³ of glucose, **G**
- 6 cm³ of methylene blue, **M**

Proceed as follows:

1. Label 5 boiling tubes 1, 2, 3, 4, 5. Place 5 cm³ of suspension **Y** into each of the boiling tubes. Ensure **Y** is well-suspended. Place all the boiling tubes into a water bath of 60°C for 5 minutes.
2. While waiting, carry out a dilution to make up 5 cm³ of different concentrations of ethanol solutions using the vials provided. When preparing the solutions, add distilled water first before adding ethanol. This is to ensure better mixing of ethanol and distilled water.

Complete Table 1.1 to show how you will make the different concentrations of ethanol solution.
[2]

Table 1.1

vial	concentration of ethanol / %	volume of distilled water / cm ³	volume of ethanol / cm ³
1	100		
2			
3			
4			
5			

Read through steps **3** to **10** to prepare a table to record your results in **(b)**, before starting the investigation.

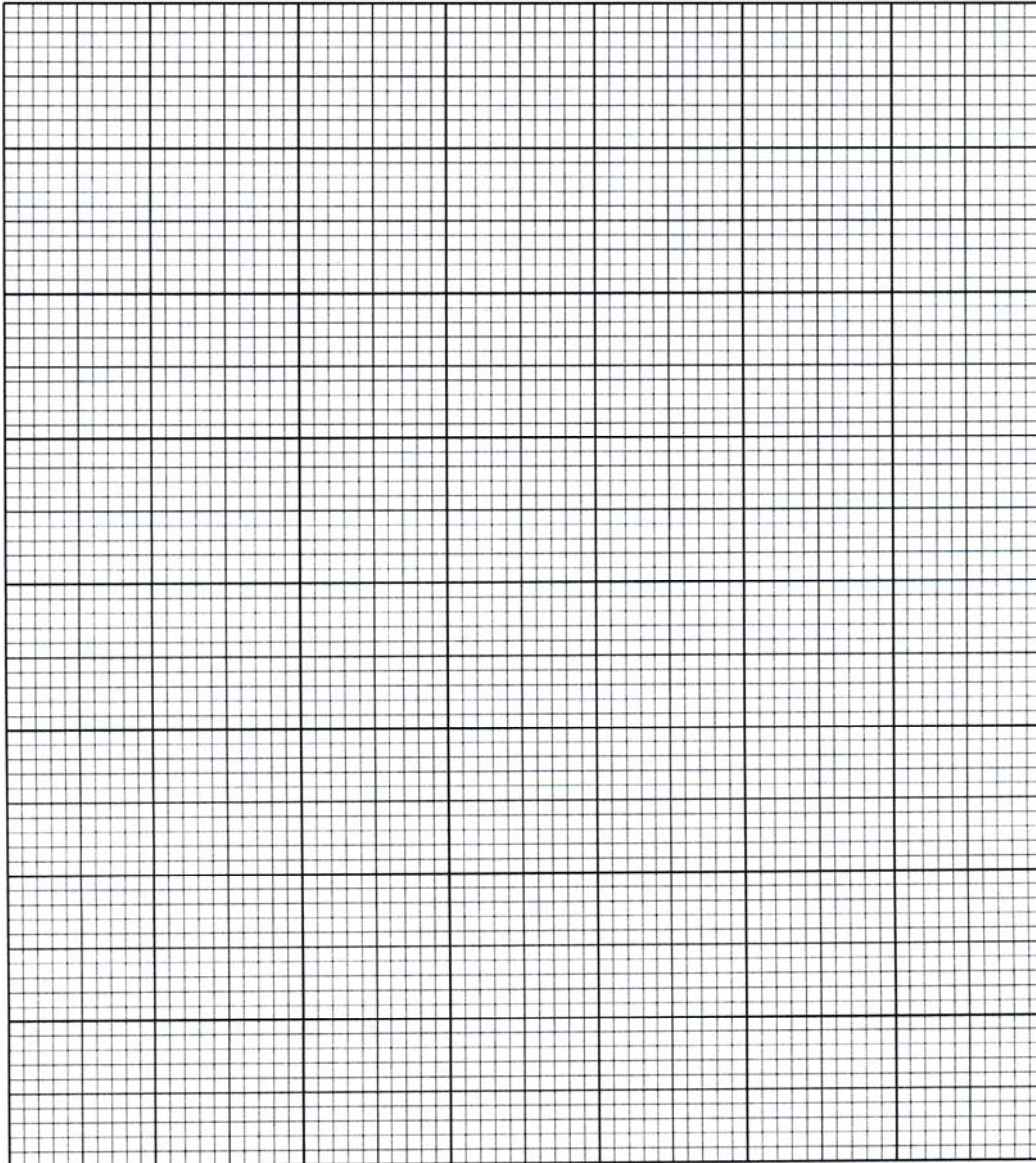
3. After 5 minutes, lower the temperature of the water bath to 45°C. Leave the boiling tubes in the water bath for 1 minute.
4. Place 2 cm³ of the ethanol solution from vial 1 into boiling tube 1 then add in 2 cm³ of **G**.
5. Use a dropper to add 1 cm³ of **M** into boiling tube 1 immediately.
6. Shake the boiling tube sufficiently to mix the contents well. The mixture should turn pale blue.
7. Carefully place the boiling tube back into the water bath and start the stopwatch. Do not shake or stir the boiling tubes from this point onwards as it may affect the results.
8. Stop the stopwatch when the blue mixture has been decolourised. The surface of the mixture in the boiling tube may remain blue. Record the time taken and hence the rate of respiration in the table prepared in **(b)**. If the mixture does not decolourise within 12 minutes, record 'more than 720' as the time taken and the rate as '0'.
9. Repeat steps 4 to 8 with boiling tubes 2, 3, 4, and 5, in turn.
10. After completing the experiment, shake boiling tube 5 vigorously about 10 times. Record your observations in part **(d)**.

(b) Record your results for each concentration of ethanol in a suitable format in the space below.

[4]

(i) Use the grid below to display your results from (b).

[4]



(ii) Discuss what these results suggest about the relationship predicted in part **(a)**. [2]

(c) State and explain your observations from step 10. [1]

(d) Suggest how adding glucose solution to the mixture increases the validity of the results. [2]

(e) One way to increase confidence in the conclusions of this investigation would be to repeat the experiment several times.

Describe two other modifications to the method that would increase confidence in the conclusions, and explain how these modifications would achieve this. [2]

[Total: 21]

2 Section A

You are required to carry out an investigation to estimate the water potential (ψ) of the cells of the plant material with which you have been provided.

You are provided with stems of a plant sample and different concentrations of sucrose solution.

Proceed as follows:

1. Using a sharp scalpel, cut a 5 cm long, straight piece, from near the middle region, of one of the specimens provided. Hold this piece of plant in a vertical position and cut it longitudinally downwards for a distance of approximately 4 cm (Fig. 2.1).

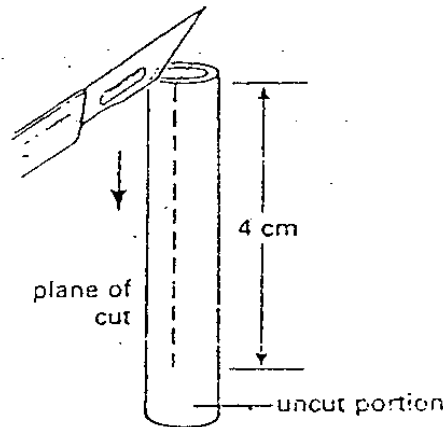


Fig. 2.1

2. You should find that the specimen has curved as shown in Fig. 2.2. Check that the distance A, between the cut pieces is at least 1 cm. If not, repeat the procedure using another specimen. Place the piece of plant tissue horizontally in the base of a clean and dry petri dish. Taking care not to squash the plant material, gently but firmly fix it to the dish using a small roll of plasticine, which you press down at X and Y (Fig. 2.2).

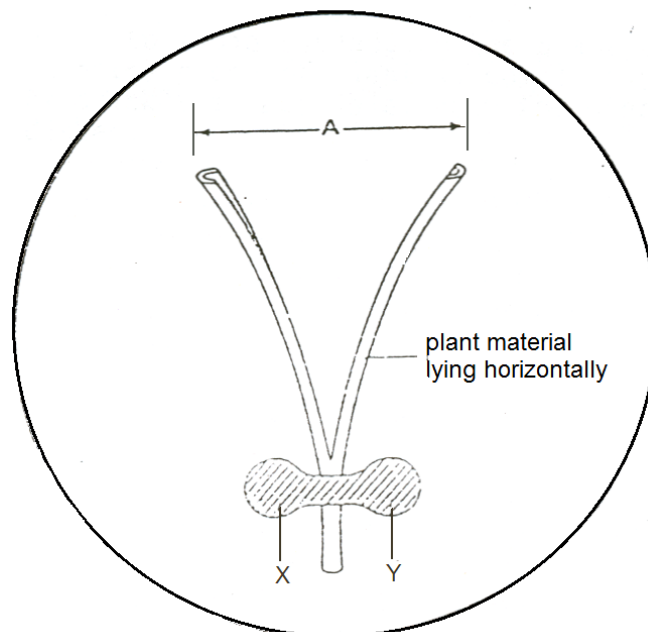


Fig. 2.2

3. Prepare **three** further dishes, using 5 cm long pieces of tissue, cut from roughly corresponding positions of three other stalks. Label your dishes 1, 2, 3 and 4.
4. Place the four dishes on the separate sheet of graph paper provided and measure, to the nearest millimeter, the distance A in each dish. Record these observations in the table below. [2]

	Dish 1 (for S1)	Dish 2 (for S2)	Dish 3 (for S3)	Dish 4 (for S4)
Initial value of A / mm				
Value of A after 10 minutes / mm				
Difference between initial and final value of A / mm				
Change / %				

5. You have been provided with the following sucrose solutions:

S1 is 0.2 mol dm^{-3}

S2 is 0.4 mol dm^{-3}

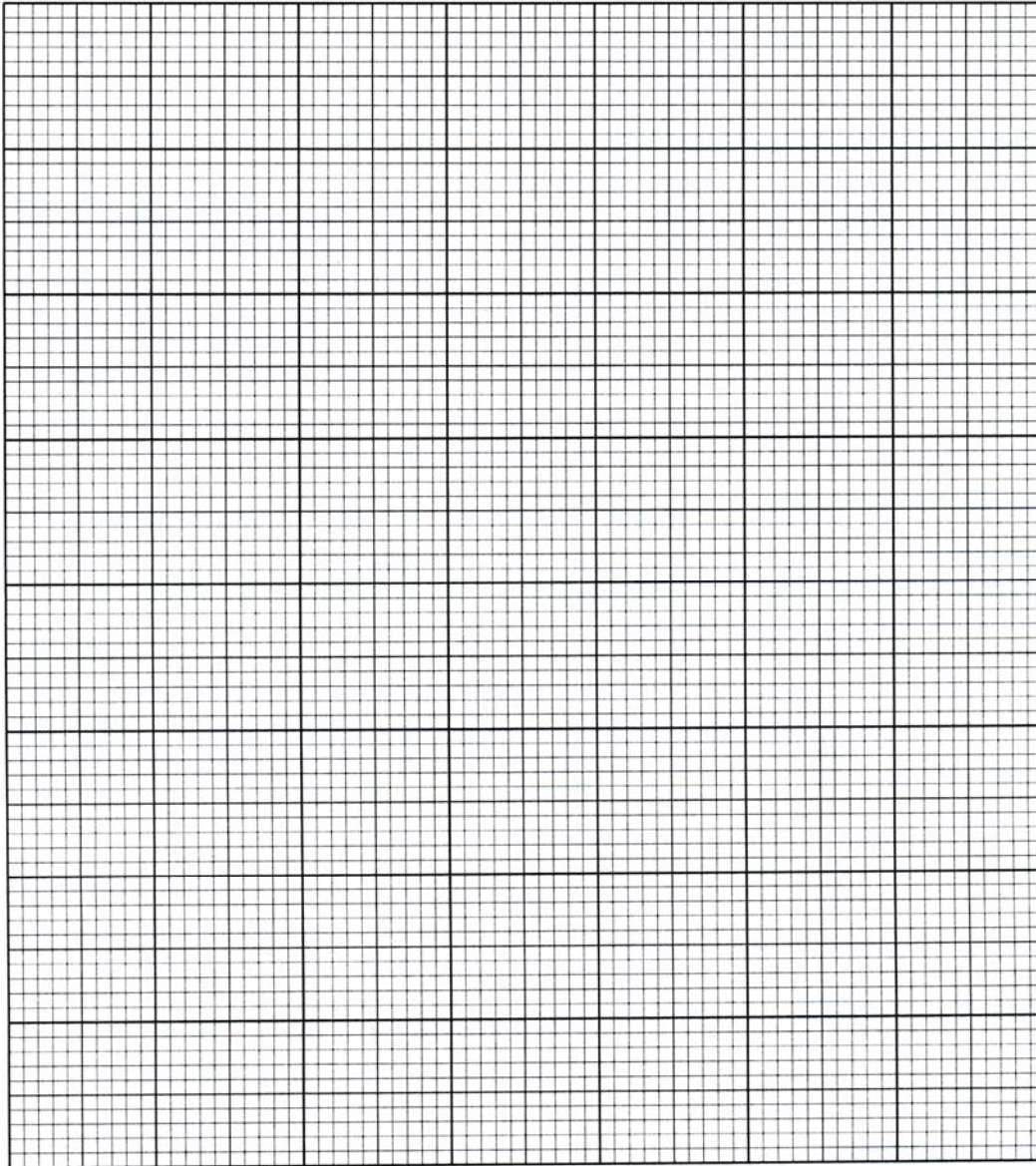
S3 is 0.6 mol dm^{-3}

S4 is 0.8 mol dm^{-3}

6. Gently, in order to avoid dislodging the plant tissue, pour **S1** into **dish 1**, so that the piece of plant is completely covered by the solution. As quickly as possible, pour the other solutions into their respective dishes.
7. Leave the dishes for 10 minutes. During this time you may begin with Section B.
8. After 10 minutes, measure (to the nearest mm) the distance A in each of the dishes. Record these measurements in the table in Step 4.
9. For each dish, calculate the percentage change in A, and also record this in the table (Step 4). State, in each case, if the value is positive or negative.

10. Plot a graph of the percentage change in A against molarity of sucrose solution.

[3]



11. The following table shows the solute potentials (ψ_s) of different concentrations of sucrose solutions, at the approximate temperature at which you have been working.

Concentration / mol dm ⁻³	Solute potential (ψ_s) / kilopascals
0.1	-260
0.2	-540
0.3	-820
0.4	-1120
0.5	-1450
0.6	-1700
0.7	-2170
0.8	-2580

12. Use the graph you have drawn, and the table above, to estimate the solute potential of the cells of this plant material. Explain fully how you arrived at your answer. [3]

Answer:

Explanation:

.....

.....

.....

.....

Section B

In this section, you will require access to a microscope and slide **K1**.

K1 is a stained, longitudinal section of a young onion root tip in which some cells are undergoing mitosis. Fig. 2.3 shows a plan diagram of **K1**.

Examine **K1** carefully, in the region labelled **A** in Fig. 2.3, using low- and high- power objectives of your microscope.

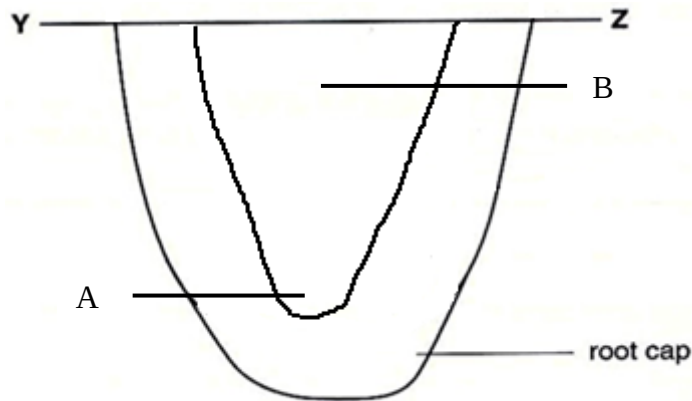


Fig. 2.3

1. Make a labelled, high-power drawing of a cell in anaphase from region **A**.

[4]

2. Using the eyepiece graticule fitted in the eyepiece lens of your microscope, and the stage micrometer, find the actual length, in μm , of the cell that you have drawn.

Show the measurements that you made and your working.

[3]

Length of cell = _____ μm

3. Measure and calculate the average length of the cells from both regions A and B. Record your results and measurements in a suitable table below.

[2]

4. Decide a statistical test that you can use to determine if there is a significant difference between the length of the cells in regions A and B.

[1]

5. A student made some measurements of the length of cells of a garlic root tip undergoing mitosis in regions A and B. A summary of the student's results is shown in Table 2.1.

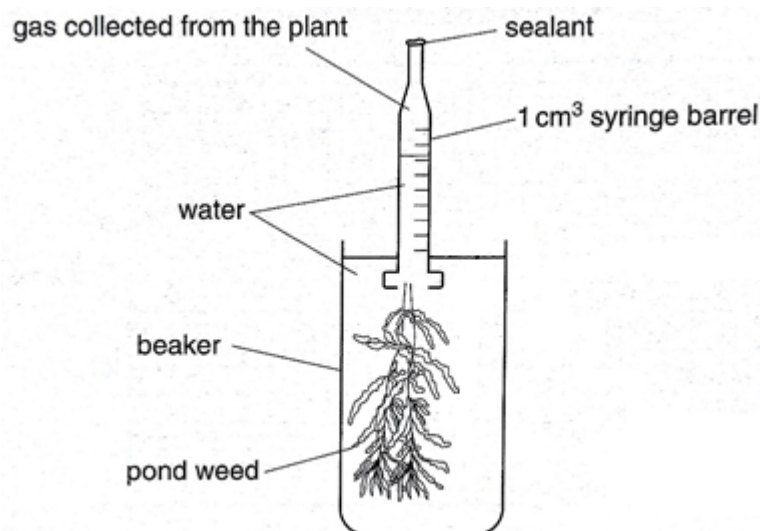
Table 2.1

Length of Cells / μm		Significance of difference
Region A	Region B	
32	58	$P < 0.05$

Comment on what these results show and suggest an explanation for any pattern. [2]

[Total: 20]

- 3 The rate of photosynthesis can either be measured by the rate at which carbon dioxide is taken in or the amount of oxygen that is given out. Some water plants release bubbles of gas from a freshly cut stem when illuminated. Light intensity is controlled using five filters, F1, F2, F3, F4, F5.



Different water plants are adapted to different light intensities. A sun-loving water plant is adapted to high light intensities while a shade-loving water plant is adapted to low light intensities.

Using this information, the set-up above and your own knowledge, design an experiment to investigate the effect of light intensity on photosynthesis in sun and shade plants.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you **must** use:

- Sun plant
- Shade plant
- Bench lamp with 60 W bulb
- 5 filters (F1, F2, F3, F4, F5) which can be adjusted to allow different amounts of light to pass through
- 1% sodium hydrogencarbonate solution

You may select from the following apparatus and use appropriate additional apparatus:

- Normal laboratory glassware, e.g. a variety of different sized beakers, measuring cylinders, and syringes for measuring volumes
- Forceps
- Timer, e.g. stopwatch

- have a clear and helpful structure such that the method you use is repeatable by anyone reading it
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[illegible]

ACJC

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

For
Examiner's
Use